YEAR 8 MATHS · CHAPTER 3 · MIND MAP

Fractions, decimals and percentages

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1 Equivalent fractions

Multiply or divide the top and bottom by the same number.

$\frac{2}{3} = \frac{8}{12}$ $\frac{18}{24} = \frac{3}{4}$

2 Operations with fractions

Add or subtract: common denominator first. Divide: flip the second fraction and multiply.

$\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$

$1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$ $\frac{3}{4} \div \frac{1}{2} = \frac{3}{2}$

3 Negative fractions

The same sign rules as integers.

$-\frac{1}{2} + \frac{1}{3} = -\frac{3}{6} + \frac{2}{6} = -\frac{1}{6}$

$-\frac{2}{3} \times \left(-\frac{3}{4}\right) = \frac{1}{2}$

4 Understanding decimals

Compare place by place, starting from the left.

0.305 is 3 tenths and 5 thousandths

0.5 is bigger than 0.45: 5 tenths beats 4 tenths

Watch out: more digits doesn't mean bigger.

5 Operations with decimals

Line up the decimal points to add or subtract. To multiply, count the decimal places. To divide, make the divisor whole.

$2.75 + 1.4 = 4.15$

$0.3 \times 0.2 = 0.06$ (two decimal places)

$4.8 \div 0.6 = 48 \div 6 = 8$

6 Recurring decimals and rounding

Divide the top by the bottom. It either stops (terminating) or repeats (recurring).

$\frac{3}{8} = 0.375$ $\frac{1}{3} = 0.\dot{3}$ $\frac{2}{11} = 0.\dot{1}\dot{8}$

2.467 to 1 decimal place is **2.5**

7 Converting

Fraction to decimal: divide. Decimal to percentage: multiply by 100.

$\frac{3}{5} = 0.6 = 60\%$

$\frac{1}{8} = 0.125 = 12.5\%$

$7\% = 0.07 = \frac{7}{100}$

8 Finding percentages

Percentage of an amount: multiply. Express as a percentage: divide and multiply by 100.

25% of 80 is **20**

18 out of 40: $\frac{18}{40} \times 100\% = 45\%$

9 Increasing and decreasing

Use a multiplier: 100% plus the increase, or 100% minus the decrease.

Up 15%: $60 \times 1.15 = 69$

Down 20%: $45 \times 0.8 = 36$

10 Percentage change, profit and loss

$$\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100\%$$

Bought for \$50, sold for \$65: profit \$15, $\frac{15}{50} = 30\%$

11 The unitary method

Find what 1% is worth, then multiply to get what you need.

15% of an amount is \$60

1% is $60 \div 15 = \$4$, so 100% is $\$400$

Check: 15% of \$400 is \$60 ✓